

PRASANNA RAJ R

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building Agentic AI systems, RAG pipelines, and LLM-powered applications using LangChain, LangGraph, FastAPI, and Python. Developed and deployed production-ready solutions including a live AI agent processing 50+ query types in under 2 seconds and a full-stack PWA with role-based access control and AI-powered recommendations. 3x National Hackathon Finalist passionate about Generative AI, backend engineering, and scalable intelligent systems.

TECHNICAL SKILLS

Programming Languages: Python, SQL, TypeScript, JavaScript

AI & Machine Learning: LangChain, LangGraph, RAG, Agentic AI, Multi-Agent Systems, Prompt Engineering, NLP, Transformers, XGBoost, Deep Learning

LLM Platforms & Tooling: Gemini API, OpenAI API, Anthropic API, Tool/Function Calling, Vector Databases (ChromaDB)

Backend: FastAPI, REST APIs, Async Programming, SQLite, Convex, Docker

Frontend: React, TypeScript, Vite, Tailwind CSS, Framer Motion

Tools & Visualization: Git, GitHub, n8n, Power BI, Pandas, Matplotlib, Tableau

PROJECTS

AgentDB — Agentic AI Database Assistant

[AgentDB](#)

- Architected a LangGraph ReAct pipeline with 5 custom LangChain tools (schema reader, SQL executor, chart generator, diagram renderer, insight generator) enabling autonomous, step-by-step reasoning over any SQLite database.
- Converted plain-English questions into instant SQL results, interactive charts, and ER diagrams, cutting response time from hours to under 2 seconds; validated with 50+ queries at iTech AI Innovation Hackathon 2026.
- Shipped full-stack app (FastAPI + React/Vite) via Docker on Render and Vercel with SSE streaming and persistent query history. *Tech: LangGraph, LangChain, Gemini 2.5 Flash, FastAPI, SQLite, React, Docker.*

Campus Connect — AI-Powered Event Management Platform (PWA)

[Campusconnect](#)

- Built a full-stack PWA for campus event management with role-based access control, QR-based ticketing/attendance, and real-time organizer analytics using React, TypeScript, and a Convex serverless backend.
- Implemented AI-powered event recommendations (Google Generative AI), OTP authentication, and ownership verification across all mutations, documented in a complete security audit.
- Configured as an installable PWA (Vite PWA Plugin + Workbox) with offline support, background sync, and auto-updates across Android, iOS, and desktop. *Tech: React, TypeScript, Vite, Tailwind CSS, Convex, Google Generative AI.*

UrbanBloom AI — Intelligent Gardening Assistant (OpenAI API + IoT)

- Built an AI gardening assistant using the OpenAI API with few-shot and system-prompt engineering, handling 10+ plant-care scenarios with personalized, context-aware recommendations.
- Integrated OpenWeather API to dynamically adapt irrigation logic to real-time weather; wired up Arduino-based sensor and relay control, eliminating manual watering.

EXPERIENCE

Data Analyst Intern — NoviTech R&D Private Limited

Sep–Oct 2024 | Bengaluru, India

- Analyzed 5+ real-time business datasets (Python, Pandas, Matplotlib, Excel), surfacing 3 key trend insights that directly informed stakeholder decisions.
- Streamlined reusable data cleaning and visualization pipelines, cutting preprocessing time by 40% for recurring reporting cycles.

Green Intern — 1M1B (1 Million for 1 Billion)

Sep–Oct 2025 | Remote, India

- Contributed data-driven research to sustainability-focused technology projects supporting initiatives serving 1M+ beneficiaries; strengthened cross-functional stakeholder collaboration.

ACHIEVEMENTS

- 3x National Hackathon Finalist** — Smart India Hackathon 2024, Advaya 2K25, Vihansa 2025.
- iTech AI Innovation Hackathon 2026** — Architected a full-stack AI agent system end-to-end within the hackathon timeline.

EDUCATION

B.E. in Artificial Intelligence and Machine Learning — Sri Sairam College of Engineering Aug 2023 – Aug 2027

CGPA: 7.89 / 10 | Currently in 6th Semester

CERTIFICATIONS

- Data Visualization: Empowering Business with Effective Insights** — Tata Group / Forage, Feb 2025
- Academy Accreditation – Databricks Fundamentals** — Databricks, Feb 2025
- Artificial Intelligence Fundamentals** — IBM, Jan 2025

LEADERSHIP & ACTIVITIES

- MAGIC Team Member, MAPP Club** — Sri Sairam College of Engineering (Aug 2024–Present); led peer-learning initiatives for 50+ club members.
- Student Coordinator, AI Club** — Sri Sairam College of Engineering (Aug 2023–Present); organized an AI quiz for 100+ students and facilitated 4+ ML/DL knowledge-sharing sessions.